Chen et al. 2020 — Nasopharyngeal Carcinoma (HNSCC)

10x scRNA-seq · 14 samples · 3,674 malignant cells retained (of 3,763; 8 samples ≥107 cells)

Recapitulation of the embryonic EMT manifold (Gastrulation / pre-Somitic / Neural-Crest programs)

GATES: G1 mixing+continuity = GO G2 module geography = GO (non-branching) G3 cross-repression = NA

Gate 1 (01→02) GO — mixing & continuity.

Harmony on sample (θ=4, n_pcs=19): iLISI 1.08→2.33 of a 3.23 ceiling (72%; ceiling <8 because P11+P01=72% of cells).

Single connected kNN component over all 3,674 cells — continuous, sample-integrated. No patient islands.

Gate 2 (03→04) GO, NON-BRANCHING — Gast/pSom overlap, Crest inverse terminal gradient.

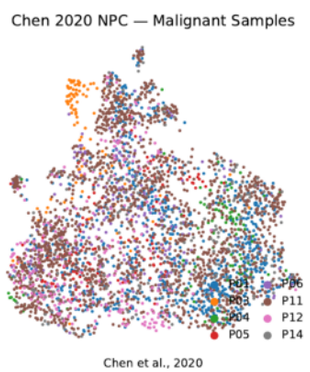
Per-node-cluster corr: Gast~pSom = +0.13 (OVERLAP, not cross-repression); Gast~Crest = −0.82; pSom~Crest = −0.50.

Curve pseudotime: corr(t,Gast)=−0.58, corr(t,pSom)=−0.11, corr(t,Crest)=+0.44 — single Gast/pSom-high→Crest-high axis.

Cycling confound cleared (Kürten lesson): Gast~G2M = −0.49 — progenitor is quiescent, not proliferation-inflated.

Gate 3 (04→05) NA — no bifurcation; branching-only correlation/modules notebooks not produced (spec §6).

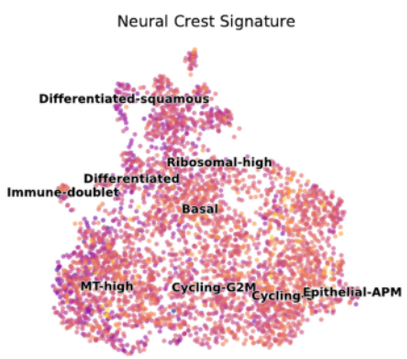
01 · sample-mixed manifold



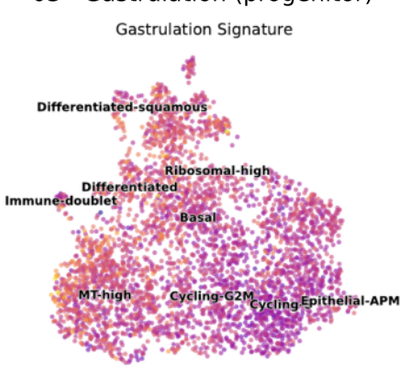
02 · curve trajectory (pseudotime)



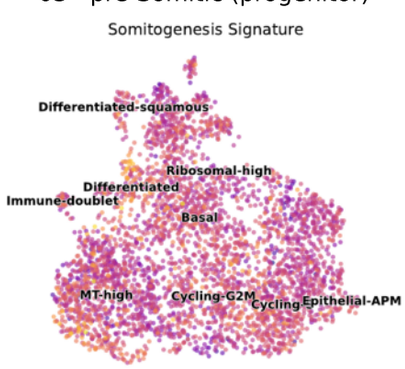
03 · Neural-Crest (terminal)



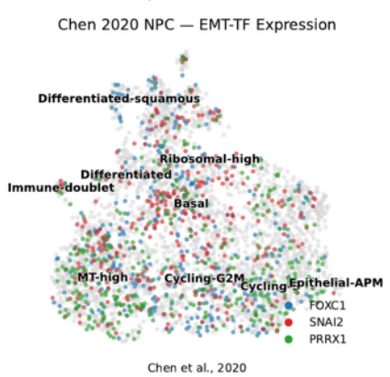
03 · Gastrulation (progenitor)



03 · pre-Somitic (progenitor)



04 · EMT-TFs (FOXC1/SNAI2/PRRX1)



VERDICT — the embryonic pattern reproduces as the NON-BRANCHING case: a single developmental axis from a Gast/pSom-high progenitor to a Crest-high terminus (structurally the Kürten 2021 outcome, not Puram's branched hub-and-spoke). Detectable EMT-TFs are FOXC1, SNAI2, PRRX1, ZEB2 (canonical SNAI1/ZEB1/TWIST below detection).

Key choices (data-derived): min-cell threshold ≥107 (count-gap rule, 8/14 samples) · no sc.pp.scale · 1,290 overdispersed genes · PCA n_pcs=19 · Harmony θ=4 · curve Nodes=10 · root = Gast/pSom-high tip · union control pool for scoring.